MODULE 1: GENERAL WILDERNESS SURVIVAL

1.1. SURVIVAL PRIORITIES

In an emergency in the wilderness, the key to survival is efficient management of the body's resources and a time-based hierarchy of actions. The human body is governed by strict physiological and thermodynamic laws. A standard operational rule used by rescue services and the military is the Rule of Threes.

Limit	Without	Main risk
3 minutes	Oxygen	Suffocation, cerebral hypoxia, drowning
3 hours	Shelter	Extreme conditions: hypothermia or heat stroke
3 days	Water	Dehydration, declining organ function
3 weeks	Food	Starvation, exhaustion, autophagy

In practice, survival priorities must be dynamically adapted to the environment. In a cool temperate or winter climate, after immediate safety has been secured, the most important actions are building shelter and starting a fire to stop heat loss.

Survival Psychology: S.U.R.V.I.V.A.L.

A major factor that can incapacitate a survivor in the first hours is panic, which leads to irrational decisions and rapid depletion of glycogen reserves. The US Army uses the following procedural acronym:

- **S - Size up the situation:** assess your location, health, and equipment.
- **U - Undue haste makes waste:** act without unnecessary haste and avoid injury.
- **R - Remember where you are:** keep track of your position and terrain.
- **V - Vanquish fear and panic:** control fear, panic, and breathing.
- **I - Improvise:** use available materials in new ways.
- **V - Value living:** maintain the will to live.
- **A - Act like the natives:** observe animals, local people, and nature.
- **L - Learn basic skills:** rely on fundamental survival skills.

1.2. SHELTER

Shelter creates a microclimate separating the body from wind, precipitation, UV radiation and, most importantly, frozen ground. Heat is lost through four physical mechanisms:

- **Conduction:** direct heat exchange with the ground. The source states that ground can remove heat from the body more than 20 times faster than air; an insulating mat or branch floor is therefore critical.
- **Convection:** heat exchange caused by moving air. Wind greatly increases perceived cold.
- **Radiation:** emission of infrared energy from the body to the surroundings.
- **Evaporation:** heat loss through sweating and breathing; wet clothing conducts heat much faster than dry clothing.

Types of Improvised Shelters

Shelter	Time / materials	Characteristics
Lean-to	1.5-2 h / horizontal ridgepole, supports, branches	Protection from wind from one direction; open front; useful with a fire reflector. Low-to-medium insulation.
A-frame	2-3 h / ridgepole, rafters at about 45°, leaf or spruce-bough covering	Good rain and wind protection; low internal volume limits convective heat loss. Covering should be about 30-50 cm thick.

Shelter	Time / materials	Characteristics
Snow cave / Quinzee	3-5 h / piled and compacted snow; allow about 2 h to crystallize	Excellent winter insulation. Interior may remain around 0 to +2°C even at -30°C outside. Requires ventilation.
Improvised tarp / poncho shelter	15-30 min / 3x3 m tarp, cord, stakes	Rapid rain protection but little thermal insulation; requires a separate raised sleeping platform.

GROUND INSULATION / BEDDING RULE

Never lie directly on the ground. Build an insulating mattress from roughly 20-30 cm of dry conifer boughs, leaves, or moss. After compression under body weight, the layer should still be at least about 15 cm thick.

1.3. FIRE

In survival, fire provides warmth, dries clothing, cooks food, treats water, raises morale, and serves as a signal. Combustion requires the fire triangle: fuel + oxygen + temperature (heat).

Fuel classification

- **Tinder:** very low ignition point and high surface area; examples include birch bark, birch fungus, char cloth, thistle/cattail fluff, and feather sticks.
- **Kindling:** small, completely dry sticks ranging from matchstick to pencil thickness.
- **Fuel wood:** arm-thick and larger wood. Hardwoods provide long-lasting embers; softwoods burn faster with higher flames and more smoke.

Fire-lay geometry

- **Teepee:** cone-shaped, excellent airflow, quick ignition and intense heat, but consumes wood rapidly.
- **Log cabin:** alternating square layers of wood; stable with room for embers and suitable for cooking.
- **Long / Siberian fire:** two thick logs parallel with embers between them, producing even radiant heat along a sleeping area.
- **Reflector fire:** a fire in front of a vertical wall of stones or logs that reflects infrared heat toward the shelter.

1.4. WATER

The source gives an adult resting water requirement of about 2.5-3 litres per day, increasing to 5-8 litres during physical exertion or high temperatures.

- **1-5% body-mass loss:** thirst-related symptoms, irritability, dark urine, and reduced performance.
- **6-10%:** dizziness, headache, dry mouth/lack of saliva, speech impairment, bluish extremities.
- **>11%:** delirium, swollen tongue, coma, kidney failure, and death.

1.5. SIGNALING

Signaling aims to shorten search-and-rescue operations. Signals can be visual, acoustic, or electronic.

- **Rule of Three:** any signal repeated three times is treated as an international distress signal - for example three shots, three loud whistle blasts, or three signal fires arranged in a triangle.
- **Signal mirror / heliograph:** directs a bright flash toward rescuers or aircraft.
- **Smoke fires:** fresh green branches, moss, or wet grass can create dense daytime smoke.

Ground-to-air symbol	Meaning
V	Require assistance
X	Require medical assistance
N	No / Negative

Ground-to-air symbol	Meaning
Y	Yes / Affirmative
↑	Proceeding in this direction

1.6. ORIENTATION AND NAVIGATION

Field navigation can rely on instruments such as a compass or on natural indicators.

- **Shadow-stick method:** place a vertical stick in the ground, mark the end of its shadow, wait 15-20 minutes, and mark the new shadow tip. The line between the marks approximates the east-west axis.
- **Polaris:** in the Northern Hemisphere, the North Star indicates true north. The source describes locating it using the Big Dipper.
- **Vegetation and natural signs:** the manual mentions bark, moss/lichen growth, and tree-ring asymmetry as rough indicators; these should be treated as environmental clues rather than precise navigation instruments.